

# Syed Nahin Hossain

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Portfolio: nahin-jzs.github.io | Google Scholar | GitHub: NAHIN-JZS | LinkedIn

## Research Interests

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3D Computer Vision | 3D Scene Understanding | Computer Vision | Deep Learning | Machine Learning

## Education

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### Bachelor of Science in Computer Science and Engineering

Khulna University of Engineering & Technology, Khulna, Bangladesh

CGPA – 3.50/4.00

Graduated: May 2023

- **Thesis:** Gray Scale to Color Image Conversion using Generative Adversarial Network

**Supervisor:** Dr. Sk. Md. Masudul Ahsan

#### Findings:

1. Conditional GAN excels at generating realistic color photos, with training in Lab and YCbCr color spaces proving most efficient for image colorization.
2. Using pre-trained models enhances the generator's efficiency, and focusing on specific tasks yields more efficient colorization.

## Research and Publications

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### Research

1. "Monocular 3D Vision for Vehicle Distance Estimation Toward Digital Twin Traffic Systems."  
Status – Ongoing work
2. "Gray Scale to Color Image Conversion using Generative Adversarial Network."  
Status – Submitted in Q1 Journal
3. "Bone Fracture Detection and Localization Using Transfer Learning Techniques on a Comprehensive Dataset."  
Status – Manuscript Ready
4. "Advanced Modeling of UHPC Compressive Strength: Insights from Eleven Machine Learning Algorithms."  
Status – Manuscript Ready

### Journal Publications

1. "Advancing Water Quality Management: Comparative Analysis of Machine Learning Models for Predicting WQI in Dhaka's Rivers." Nishat, M. H., Khan, M. H. R. B., Ahmed, T., **Hossain, S. N.**, Ahsan, A., El-Sergany, M. M., Shafiquzzaman, M., Imteaz, M. A., & Alresheedi, M. T. *Environmental Sciences Europe* (2025). **IF: 6.0** | **Citations: 67** | DOI: 10.1186/s12302-025-01078-w

### Conference Publications

1. "Automatic License Plate Recognition System for Bangladeshi Vehicles Using Deep Neural Network." **Hossain, S. N.**, Hassan, M. Z., & Al Masba, M. M. *Proc. Int. Conf. on Big Data, IoT and Machine Learning (BIM)*, Springer (2021). **Best Paper Award** | **Citations: 22** | DOI: 10.1007/978-981-16-6636-0\_8
2. "Bangladeshi Local Vehicle Recognition with A Comprehensive Dataset using Transfer Learning Techniques." Rahman, M. S., Hassan, M. Z., **Hossain, S. N.**, Masrur, N., & Rabbi, J. *IEEE* (2022). **Citations: 8** | DOI: 10.1109/STI56238.2022.10103325
3. "Transfer Learning for Vehicle Recognition: A Comprehensive Dataset and Evaluation on Bangladeshi Roads." Prapty, A. S., **Hossain, S. N.**, Efti, M. S. P., Rahman, M. M., & Nurullah, M. *IEEE* (2026). **IEEE Xplore: 11545960**

## Work Experience

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### Lecturer, Department of Computer Science & Engineering

Bangladesh Army University of Science & Technology, Khulna, Bangladesh

July 2025 – Present

Full Time

### Lecturer, Department of Computer Science & Engineering

Bangladesh Army University of Engineering & Technology, Natore, Bangladesh

July 2023 – July 2025

Full Time

## Technical Projects

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|-----------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Automatic License Plate Recognizer</b>                                                                                               | 2021        |
| <ul style="list-style-type: none"><li>• <i>Technologies:</i> Python, Transfer Learning , YOLOv4, Tensorflow model Zoo, OpenCV</li></ul> |             |
| <b>Virtual Makeup Try-On System</b>                                                                                                     | 2020 – 2021 |
| <ul style="list-style-type: none"><li>• <i>Technologies:</i> Python, OpenCV, Deep Learning</li></ul>                                    |             |

## Technical Skills

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**Programming Languages:** Python, C++, C, JavaScript, HTML  
**Frameworks & Libraries:** TensorFlow, PyTorch, OpenCV, OpenGL  
**Tools:** Git, Latex, Anaconda, VS Code

## Online Courses & Certifications

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|--------------------------------------------------------------|---------------|
| • <b>Advanced Computer Vision with TensorFlow</b> – Coursera | [Certificate] |
| • <b>Deep Learning Specialization</b> – Coursera             | [Certificate] |
| • <b>Sequence Models</b> – Coursera                          | [Certificate] |
| • <b>Machine Learning for Everyone Track</b> – DataCamp      | [Certificate] |

## Awards and Honors

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|------------------------------------------------------------------------------|-------------|
| <b>Best Paper Award</b>                                                      | 2021        |
| <i>International Conference on Big Data, IoT and Machine Learning (BIM)</i>  |             |
| <b>Dean's Award</b>                                                          | 2021 – 2022 |
| <i>Khulna University of Engineering &amp; Technology, Khulna, Bangladesh</i> |             |
| <b>Board Scholarship, Higher Secondary Certificate Examination (HSC)</b>     | 2017        |
| <i>Board of Intermediate and Secondary Education, Jashore, Bangladesh</i>    |             |
| <b>Board Scholarship, Secondary School Certificate Examination (SSC)</b>     | 2015        |
| <i>Board of Intermediate and Secondary Education, Jashore, Bangladesh</i>    |             |
| <b>Board Scholarship, Junior School Certificate Examination (JSC)</b>        | 2012        |
| <i>Board of Intermediate and Secondary Education, Jashore, Bangladesh</i>    |             |

## Leadership and Affiliation

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| <b>Assistant Hall Provost</b>                                    | 2025 – Present |
| <i>Student Hall-2, BAUST, Khulna, Bangladesh</i>                 |                |
| <b>Senior Instructor</b>                                         | 2025           |
| <i>Al Biruni Society of Scientific Studies</i>                   |                |
| <b>President</b>                                                 | 2022 – 2023    |
| <i>Jashore Association of KUET (JAK)</i>                         |                |
| <b>Senior Patrol Leader</b>                                      | 2013 – 2015    |
| <i>1 no Jashore Zilla School Scout Team, Jashore, Bangladesh</i> |                |

## References

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1. **Dr. Sk. Md. Masudul Ahsan**  
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2. **Jakaria Rabbi**  
Assistant Professor, Dept. of Computer Science & Engineering, KUET, Khulna, Bangladesh  
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